# Research Report

2026-03-16

## Introduction

Electric vehicle (EV) prices vary substantially across models, even among vehicles with similar technical specifications. Battery capacity is one of the most visible EV characteristics and is often associated with both production cost and consumer value. However, vehicle price is not determined by battery capacity alone. Manufacturers differ in brand positioning, and premium brands may charge systematically higher prices than non-premium brands even for vehicles with similar specifications. Understanding this relationship is important for both consumers and manufacturers. For consumers, it helps clarify whether higher prices reflect meaningful technical improvements or brand positioning. For manufacturers, it provides insight into how battery capacity is translated into pricing across different market segments.

This report studies EV models listed for sale in Germany and investigates whether battery capacity is associated with vehicle price, and whether that association differs between premium and non-premium manufacturers. The research questions are:

1. Among electric vehicle models listed for sale in Germany, is battery capacity positively associated with vehicle price?
2. Does the association between battery capacity and price differ between premium and non-premium manufacturers?

The data come from the *Electric Vehicle Specifications and Prices* dataset obtained from Kaggle and compiled by web scraping EV Database. EV Database aggregates manufacturer-reported technical specifications and listed prices for electric vehicle models listed for sale in Germany. The Kaggle dataset page indicates that the dataset entry was updated "2 years ago," which provides some background on when the posted dataset was last refreshed. However, that platform update note should not be interpreted as a precise statement of the model year coverage or the exact scrape date of the underlying EV listings. The dataset contains 360 observations, where each row represents a distinct EV model. This dataset should not be interpreted as a complete census of all electric vehicles that have ever existed or all EVs ever sold in Germany. Instead, it is best understood as a snapshot of the EV models included in EV Database when the source was scraped and the Kaggle dataset was assembled. The exact calendar year or scrape date is not stated in the raw CSV used here, so the report avoids treating the data as representing a single fully specified year. Since the data are observational rather than experimental, all conclusions in this report are associative rather than causal.

The response variable is `Price.DE.`, the listed vehicle price in euros. The main explanatory variable is `Battery`, measured in kilowatt-hours (kWh). A second key variable is manufacturer segment, coded as premium or non-premium. Following the proposal, Tesla, BMW, Audi, Mercedes-Benz, and Porsche are classified as premium brands based on their widely recognized positioning in the European automotive market. This grouping is intended to reflect broad brand-tier differences rather than a formal regulatory classification. Other quantitative variables that may be useful during model building include `Range`, `Efficiency`, `Top_speed`, `Acceleration..0.100.`, and `Fast_charge`. No major data transformations were applied to the variables. In particular, all quantitative variables were analyzed on their original scales. The only modification was to simplify the analysis by not distinguishing between specific vehicle models, and instead grouping vehicles at the manufacturer level through the premium versus non-premium classification.

This report uses multiple linear regression because the response is continuous and the main goal is inference about average relationships. In particular, including an interaction between battery capacity and market segment allows the battery-price relationship to differ between premium and non-premium manufacturers, which directly addresses the second research question. Based on the course material, variables named in the research question remain in the model throughout model selection even if a selection procedure would otherwise remove them.

# Exploratory Analysis

## Data Preparation

The raw dataset is stored locally in the project folder as `data/EV_cars.csv`. Using a relative file path makes the analysis reproducible for all group members and avoids machine-specific paths such as `~/Desktop/...`.

The raw file contains 360 observations and 9 recorded variables: `Battery`, `Car_name`, `Car_name_link`, `Efficiency`, `Fast_charge`, `Price.DE.`, `Range`, `Top_speed`, and `acceleration..0.100.`. Because the data were scraped from an online database, the observed models reflect what EV Database included during the underlying collection period, not necessarily every EV available in the German market before or after that time. This matters for interpretation: the analysis is best understood as describing relationships among the EV models represented in this snapshot, rather than all EVs from a specific stated year. The response variable `Price.DE.` has 51 missing values, and `Fast_charge` has 2 missing values. Since price is the response, rows with missing price cannot be used for the regression analysis and are removed. The two rows with missing `Fast_charge` are retained during initial cleaning and are only excluded in models that include `Fast_charge`.

To prepare the analysis dataset, a manufacturer variable was extracted from the vehicle name, the premium/non-premium grouping from the proposal was applied, the acceleration variable was renamed to a cleaner label, and only the variables relevant to the analysis were retained. Identifier variables such as `Car_name_link` were not used as predictors because they do not have an interpretable statistical relationship with price. Manufacturer names are extracted heuristically and may introduce minor misclassification for multi-word brands.

After removing rows with missing price, the main analysis dataset contains 309 EV models. A check of duplicate `Car_name` values shows that some names appear more than once. These rows were retained because they are not exact duplicates of the full record. In several cases, the repeated names correspond to different EV Database links and slightly different technical specifications, which suggests updated entries or distinct configurations rather than accidental duplication. For example, duplicated Mercedes entries often appear as an older version with missing price and a newer version with a reported price, while the duplicated Mini Cooper SE entries differ substantially in battery capacity and range and therefore represent genuinely different configurations.

For the main regression, one practical choice is to fit models using complete cases on the variables included in each model. If `Fast_charge` remains in a candidate model, the effective sample size becomes 307 rather than 309 because of the two missing charging values. This should be noted when comparing models. The 51 models with missing prices do not appear dramatically different from the retained models in terms of battery capacity, range, efficiency, top speed, acceleration, or fast-charge capability, which reduces concern that the missing prices are concentrated only among very low-end or very high-end vehicles. However, because the missingness mechanism is not known, some selection bias remains possible.

## Exploratory Summaries

Exploratory analysis serves two purposes in this study. First, it helps describe the data to readers who are not familiar with the EV market. Second, it helps identify features of the data that affect the regression analysis, such as skewness, segment differences, and multicollinearity among technical variables.

Start with summary statistics (Table 1) for the response and candidate explanatory variables.

Table 1: Summary statistics for key quantitative variables

| N | Mean price | Median price | SD price | Mean battery | Median battery | SD battery | Mean range | Mean efficiency |
|---|---|---|---|---|---|---|---|---|
| 309 | 67264.28 | 56942 | 34508.25 | 71.08 | 70.5 | 20.65 | 369.13 | 195.41 |

The 309 EV models used in the main analysis have a mean listed price of 67,264 EUR and a median price of 56,942 EUR, suggesting some right-skewness caused by a smaller number of high-priced vehicles (Table 1). Battery capacity is more balanced, with a mean of 71.1 kWh and a median of 70.5 kWh. Clear differences also appear across market segments: premium vehicles are larger and more expensive on average, with a mean price of about 94,359 EUR and a mean battery capacity of 84.0 kWh, compared with 55,418 EUR and 65.4 kWh for non-premium models (Table 1). Overall, the summary statistics indicate that price is more variable than battery capacity and that premium status is associated with both larger batteries and higher prices. These patterns motivate examining the joint relationship between price, battery capacity, and market segment in the regression analysis.

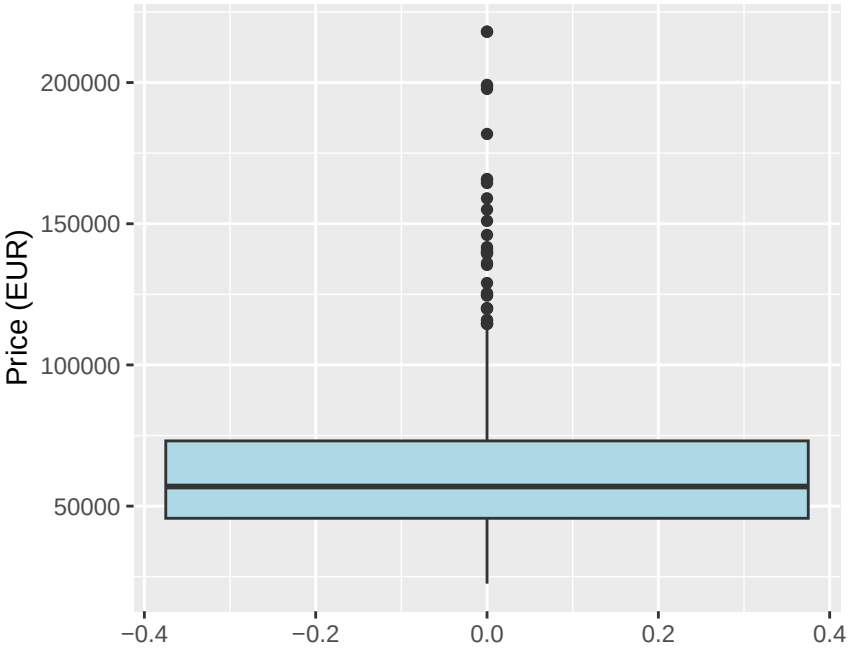


Figure 1: Boxplot of EV prices (EUR)

The boxplot of price further highlights its distributional features (Figure 1). In particular, the presence of a long upper tail and several high-price outliers confirms that price is right-skewed. Consistent with this visual evidence, the summary statistics suggest two important features that matter for modeling: price is more variable and somewhat right-skewed, while premium vehicles tend to have both larger batteries and higher prices. This motivates focusing on the joint relationship between price, battery capacity, and market segment.

The main exploratory visualization plots price against battery capacity with separate fitted lines for premium and non-premium manufacturers (Figure 2). This figure directly addresses both research questions by showing both the overall battery-price association and the extent to which that association may differ by market segment.

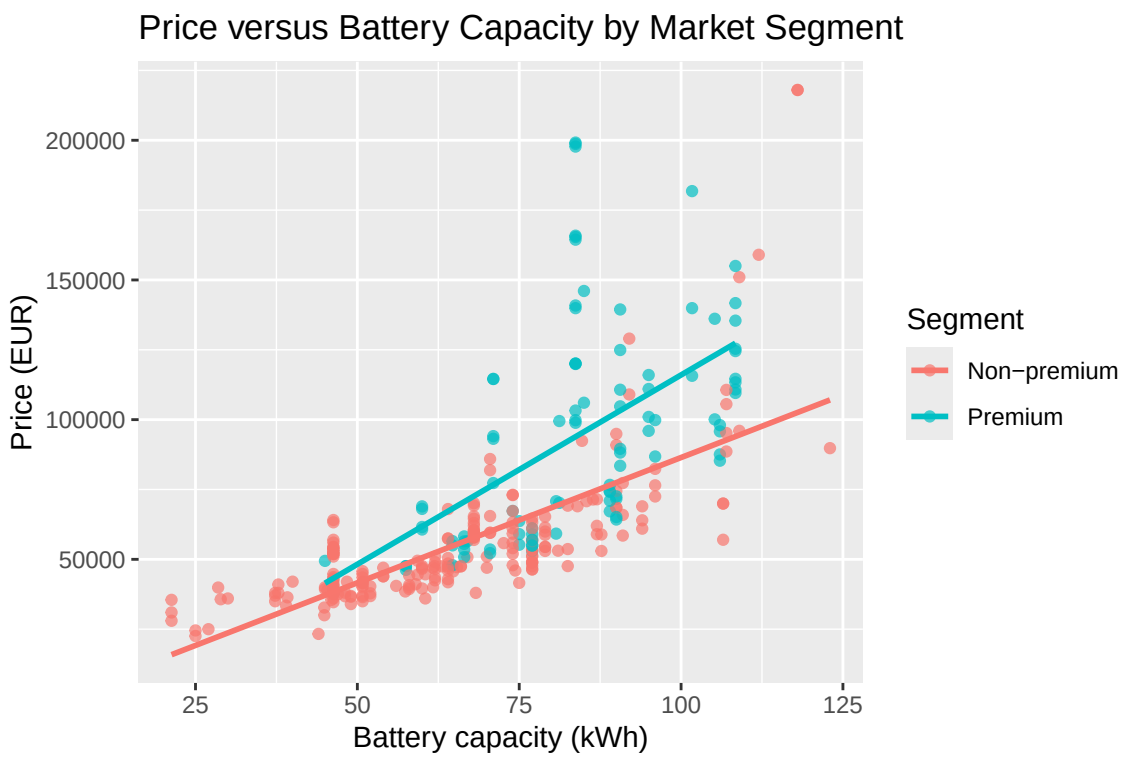


Figure 2: Price versus battery capacity by market segment

Separate simple linear regressions of price on battery capacity were fitted within each market segment, and the resulting slope estimates are reported in Table 2. For non-premium vehicles, the estimated increase in price is 896.68 EUR per additional kWh of battery capacity, with a standard error of 57.47. For premium vehicles, the corresponding estimate is substantially larger at 1,355.14 EUR per kWh, with a standard error of 222.63. In both segments, the p-values are smaller than 0.001, indicating strong statistical evidence of a positive association between battery capacity and price. Consistent with the visual pattern in Figure 2, these estimates confirm that while the relationship is positive for both segments, it is noticeably steeper among premium manufacturers. This suggests that, on average, an additional kWh of battery capacity is associated with a larger increase in listed price in the premium segment.

Table 2: Estimated battery-price relationship by market segment

| Segment | Slope estimate (EUR/kWh) | Standard error | p-value |
|---|---|---|---|
| Non-premium | 896.68 | 57.47 | <0.001 |
| Premium | 1355.14 | 222.63 | <0.001 |

Because several quantitative vehicle characteristics may be related to one another, it is also useful to examine pairwise relationships and correlations. This can help identify potential multicollinearity before model fitting.

Table 3: Correlation matrix for quantitative variables

| | Price.DE. | Battery | Range | Efficiency | Top_speed | Acceleration_0_100 | Fast_charge |
|---|---|---|---|---|---|---|---|
| Price.DE. | 1.00 | 0.70 | 0.59 | 0.17 | 0.76 | -0.53 | 0.62 |
| Battery | 0.70 | 1.00 | 0.88 | 0.16 | 0.70 | -0.61 | 0.59 |

| | Price.DE. | Battery | Range | Efficiency | Top_speed | Acceleration_0_100 | Fast_charge |
|---|---|---|---|---|---|---|---|
| Range | 0.59 | 0.88 | 1.00 | -0.31 | 0.76 | -0.75 | 0.72 |
| Efficiency | 0.17 | 0.16 | -0.31 | 1.00 | -0.21 | 0.42 | -0.29 |
| Top_speed | 0.76 | 0.70 | 0.76 | -0.21 | 1.00 | -0.84 | 0.79 |
| Acceleration_0_100 | -0.53 | -0.61 | -0.75 | 0.42 | -0.84 | 1.00 | -0.68 |
| Fast_charge | 0.62 | 0.59 | 0.72 | -0.29 | 0.79 | -0.68 | 1.00 |

Potential multicollinearity can be assessed from the correlation matrix of the main quantitative vehicle characteristics (Table 3). Several strong pairwise relationships are evident. Price is strongly positively correlated with Battery (0.70) and Top_speed (0.76), indicating that more expensive vehicles tend to have larger batteries and higher performance. At the same time, Battery and Range are themselves highly correlated (0.88), suggesting that these variables capture similar underlying information about vehicle capability. In addition, Top_speed and Acceleration_0_100 are strongly negatively correlated (-0.84), reflecting the fact that faster vehicles tend to have lower acceleration times. Overall, these correlations reveal substantial dependence among the technical variables. This suggests that including many of these predictors simultaneously may introduce multicollinearity, which can make regression coefficients unstable and harder to interpret. The broader pairwise relationships among the quantitative variables are shown in Appendix Figure 5.

# Statistical Analysis

## Modeling Strategy

The study employs a multiple linear regression framework, as the response variable—vehicle price in Euros ($Price.DE.$)—is a continuous measure. To align the statistical architecture with the core research objectives, the model is structured around a foundational set of required predictors: battery capacity ($Battery$), market classification ($market_segment$), and their corresponding interaction term. These variables are held constant throughout the refinement process to ensure that the primary research questions regarding segment-based price scaling are directly addressed. To account for potential confounding factors and technical drivers of valuation, a suite of auxiliary control variables, including $Range$, $Efficiency$, $Top_speed$, $Acceleration_0_100$, and $Fast_charge$, were initially incorporated into the full model specification.

The choice of a backward elimination strategy using partial F-tests, specifically through the drop1 procedure, was motivated by its superior ability to preserve the structural integrity of the model during the simplification phase. Unlike forward selection, which enters variables one by one and may fail to identify predictors that only become significant in the presence of others, backward elimination begins with the "full global landscape" of the data. This "top-down" methodology allows the model to account for the simultaneous influence of all candidate predictors, thereby capturing complex interdependencies and potential suppressor effects that simpler stepwise methods might overlook. By systematically pruning the least significant optional controls while shielding the required research terms from removal, this approach strikes an optimal balance between statistical parsimony and theoretical relevance. It ensures that the final model is not merely a collection of the most "statistically noisy" variables, but a refined instrument that retains only the essential technical controls necessary to isolate the true effect of battery capacity on price across different market segments.

## Initial Model

In the raw price model, the adjusted $R^2$ is approximately 0.7313. However, we must verify if the assumptions of linear regression hold before trusting these results.

## Diagnostic Discovery

A systematic review of the initial raw-price model's diagnostics revealed critical violations of Gauss-Markov assumptions:

- Heteroscedasticity: The "Residuals vs Fitted" plot displayed a prominent fan-shaped pattern, indicating that the error variance increases as the predicted price rises.
- Non-Normality: The Normal Q-Q plot exhibited a heavy upper tail, suggesting the residuals are right-skewed—a common trait in monetary datasets where high-end outliers are present.

Because these violations could lead to unstable coefficient estimates and unreliable p-values, we transition to a log-linear framework to stabilize the variance.

## Model Refinement: Log-Transformation

To address the heteroscedasticity, we applied a natural log transformation ($log(Price)$). We then re-ran our custom backward elimination procedure on the log-transformed response.

The following table presents the results of our final, statistically robust model

## Main Model Results (log linear)

The final model was fitted on 307 observations, as 2 additional rows were excluded due to missing values in the Fast_charge variable. The final regression results for the log-transformed price model are presented in Table 4. After the backward elimination procedure, Acceleration_0_100 was the only variable removed, while Range, Efficiency, Top_speed, and Fast_charge were retained as statistically significant controls. The model explains 84.12% of the variance in log-price (Adjusted $R^2 = 0.8412$).

The lack of statistical significance for the standalone Battery term in the non-premium segment ($p = 0.1517$) is likely due to its high correlation with Range ($r = 0.88$). In the log-linear model, the unique explanatory power of battery capacity is partially absorbed by the range and performance metrics, though its interaction with market segment remains a critical and significant driver of price.

Table 4: Coefficient estimates for the final regression model

| | Estimate | Std. Error | t value | Pr(>\|t\|) |
|---|---|---|---|---|
| (Intercept) | 8.0131 | 0.2184 | 36.6899 | 0.0000 |
| Battery | -0.0047 | 0.0033 | -1.4371 | 0.1517 |
| market_segmentPremium | -0.3002 | 0.1120 | -2.6802 | 0.0078 |
| Range | 0.0018 | 0.0006 | 2.8726 | 0.0044 |
| Efficiency | 0.0075 | 0.0010 | 7.4841 | 0.0000 |
| Top_speed | 0.0057 | 0.0005 | 10.5893 | 0.0000 |
| Fast_charge | 0.0003 | 0.0001 | 3.7778 | 0.0002 |
| Battery:market_segmentPremium | 0.0051 | 0.0013 | 3.7781 | 0.0002 |

### Interpretation of Results

**The Role of the Luxury Interaction Effect** The most critical finding of the study is the highly significant interaction term, $Battery : market_segmentPremium$ ($\beta = 0.0051, p < 0.001$). This coefficient provides compelling evidence that the marginal valuation of battery capacity is not uniform across the

industry but is instead contingent upon brand positioning. In the log-linear context, this positive interaction indicates that for every additional kilowatt-hour ($kWh$) of battery capacity, premium brands command a significantly higher percentage increase in price compared to non-premium brands. This suggests a "luxury multiplier" effect: in the premium segment, hardware scaling is not merely a functional upgrade but is marketed and priced as a high-value performance differentiator, allowing these manufacturers to capture greater consumer surplus from battery advancements.

**Segment-Specific Dynamics: Premium vs. Non-Premium** For the baseline non-premium segment, the model reveals a small and statistically non-significant negative coefficient for battery capacity ($\beta = -0.0047, p = 0.1517$). While a negative slope may seem counterintuitive, it signifies that once range, efficiency, and top speed are held constant, the physical size of the battery does not independently drive price for mass-market vehicles. In this segment, consumers likely view the battery as a means to an end ($Range$); if a vehicle can achieve a high range with a smaller, more efficient battery, the physical $kWh$ count offers no additional status or price premium.

Conversely, for the premium segment, the effective slope is the sum of the baseline and the interaction term (\$ -0.0047 + 0.0051 = 0.0004\$). While the net effect is near zero when all other performance metrics are fixed, the stark contrast between the two segments confirms that premium manufacturers successfully decouple battery capacity from simple utility, treating it as part of a high-performance "bundle" that scales more aggressively with price.

**The "Performance Bundle" and Control Variables** The model results for the auxiliary variables—$Range$ ($p = 0.0044$), $Top_speed$ ($p < 2 \times 10^{-16}$), and $Efficiency$ ($p < 0.001$)—further validate the hypothesis that the German EV market prices vehicles as integrated "performance bundles." The strong positive significance of $Top_speed$ and $Range$ confirms that traditional automotive value drivers (velocity and endurance) remain the dominant factors in EV valuation. The significance of $Efficiency$ is particularly noteworthy; it suggests that the market rewards engineering refinement. Vehicles that can extract more utility from their energy storage are priced higher, indicating that savvy EV consumers (or the manufacturers' pricing strategies) value technical sophistication over raw battery volume.

### Diagnostics and Model Selection

Initial diagnostics of the untransformed price model revealed significant violations of linear regression assumptions. As shown in Appendix Figure 3, the residual-versus-fitted plot exhibited a prominent fan-shaped pattern, indicating heteroscedasticity where error variance increases with the fitted values. Furthermore, the normal Q-Q plot indicated a heavy right tail, suggesting residuals were not normally distributed.

To address these issues, we adopted a log-linear model ($log(Price)$). This transformation successfully stabilized the error variance and normalized the residuals, providing a much more robust foundation for inference. The log-model also improved the fit significantly, increasing the Adjusted $R^2$ from 0.7313 to 0.8412. Consequently, the log-linear model is retained as our primary model for interpreting the battery-price relationship across different manufacturer segments.

## Discussion and Conclusion

### Discussion and Synthesis of Findings

The empirical evidence provided by this study confirms that the relationship between battery capacity and electric vehicle pricing is fundamentally moderated by market positioning. While a cursory glance at the raw data might suggest a linear scaling of costs, the refined log-linear analysis uncovers a more complex "luxury multiplier" effect that defines the premium segment. In this context, battery capacity does not

merely function as a cost-of-goods component; rather, it serves as a primary lever for value appropriation. For premium manufacturers, each additional kilowatt-hour of energy storage is leveraged to command a disproportionately higher price premium, suggesting that in the upper tiers of the German market, technical hardware is inextricably linked to brand prestige and consumer status.

### Theoretical Interpretations and Industry Implications

The transition from a standard linear model to a log-linear specification was not merely a statistical adjustment for heteroscedasticity, but a theoretical necessity to capture the multiplicative nature of automotive valuation. This shift highlights that as vehicles move up the performance ladder, price increases are compounding rather than additive. In the premium market, the "price of power" is amplified because battery size is bundled with high-performance motors, advanced thermal management, and elite brand signaling. This implies that premium brands have successfully decoupled their pricing strategies from the underlying commodity costs of lithium-ion cells, instead charging a premium for the "performance envelope" that a larger battery enables. For non-premium brands, however, the lack of a significant independent battery effect suggests a high level of price sensitivity, where consumers view battery size strictly through the lens of functional utility and range.

### Limitations and Avenues for Future Research

While the current model explains a substantial 84% of price variance, it is important to acknowledge the limitations inherent in a cross-sectional snapshot of the German EV market. The remaining unexplained variance likely resides in qualitative dimensions that are difficult to quantify, such as cabin material quality, the sophistication of proprietary software ecosystems, and autonomous driving capabilities. Furthermore, the data reflects a specific moment in the technological lifecycle of electric mobility. As global battery production scales and chemistries evolve toward solid-state or cobalt-free alternatives, the "luxury premium" identified here may face downward pressure. Future longitudinal research would be invaluable in tracking whether this brand-based price wedge narrows as battery technology becomes increasingly commodified and accessible across all market segments.

### Concluding Remarks

In conclusion, while battery capacity remains a foundational determinant of electric vehicle pricing, its economic impact is significantly magnified by the strategic tiering of the market. The findings demonstrate that the valuation of energy storage is not a fixed engineering constant but is heavily dictated by brand-specific performance expectations and market positioning. As the industry moves toward mass-adoption, understanding this interplay between hardware scaling and brand equity will be essential for both policymakers and manufacturers navigating the transition to a fully electrified automotive landscape.

## Statement of AI Use

Generative AI was used to help organize the structure and wording of this report and to assist in preparing a separate reproducible analysis script. The statistical conclusions, numerical results, and model interpretations are based on outputs generated from the dataset itself.

Representative prompts used:

1. "Help revise the writing so the report reads like a formal statistical report rather than notebook-style notes."

2. "Help update the data-preparation description so it accurately reflects the raw dataset, missing values, and derived variables used in the analysis."
3. "Help strengthen the wording in response to feedback on interaction interpretation, diagnostics, model selection, and limitations."
4. "Help move less essential material to the appendix so the main body remains concise and within the page limit."

# Appendix

## Diagnostic Plots

Supporting diagnostic plots are provided in Appendix Figure 3 and Appendix Figure 4.

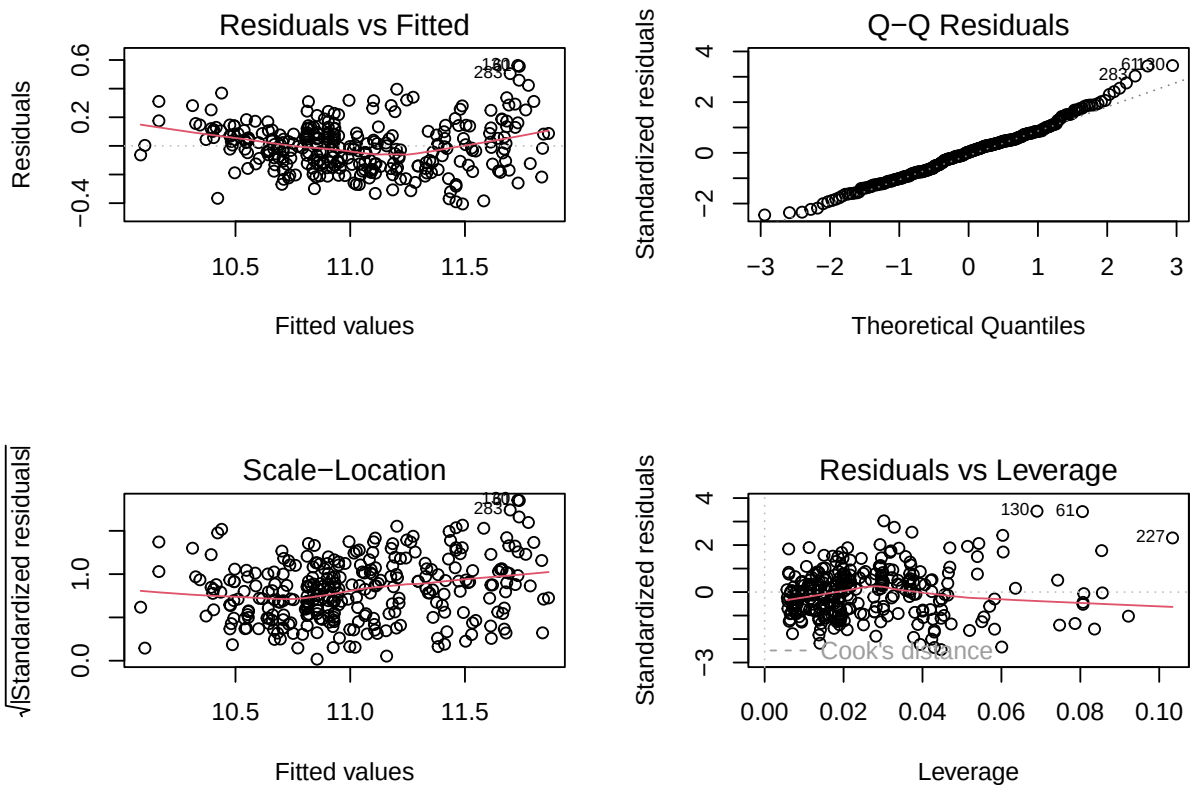


Figure 3: Diagnostic plots for the selected regression model

## Pairwise Relationships Among Quantitative Variables

An additional exploratory check of pairwise relationships among the quantitative variables is shown in Appendix Figure 5. This figure is included in the appendix rather than the main body because it is useful for supporting model-building decisions, but it is not the most direct figure for answering the research questions.

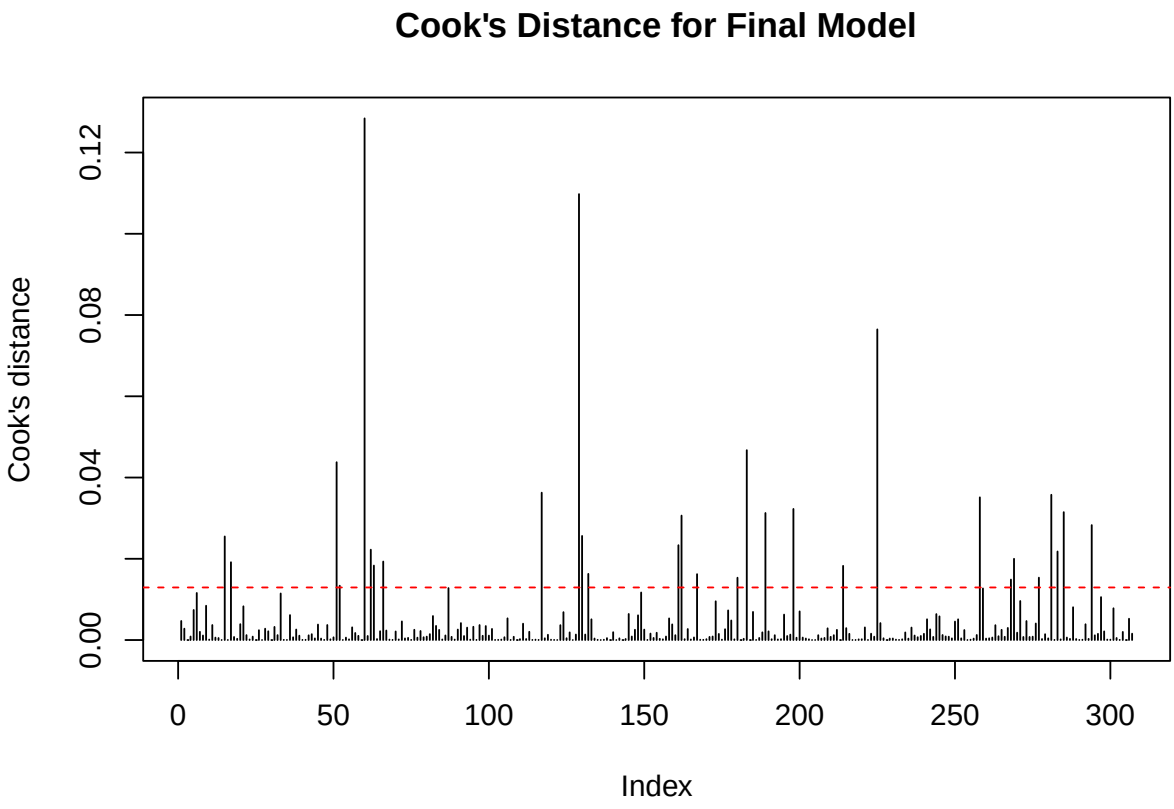


Figure 4: Cook's distance for the selected regression model

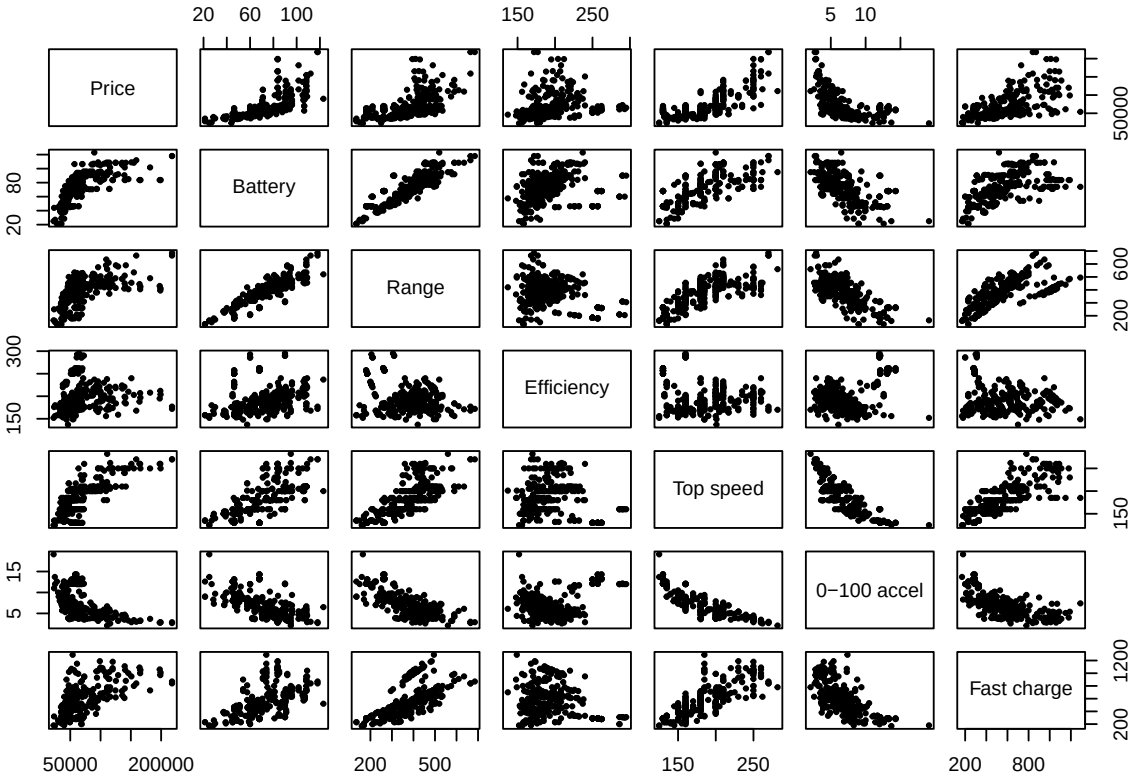


Figure 5: Scatterplot matrix for quantitative variables used in the analysis

The scatterplot matrix in Appendix Figure 5 is mainly helpful for visually checking which explanatory variables appear strongly associated with price and which predictors appear highly related to one another. In this dataset, it supports the conclusion that variables such as `Battery`, `Range`, `Top_speed`, and `Fast_charge` share substantial overlap, which helps explain why a simpler final model is preferred.